

Effect of Bypass Injection Angle and Cavity Convergence Angle on the Thrust Vectoring Performance of a Bypass Dual Throat Nozzle

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Tools: ANSYS Design Modeler, ANSYS Meshing, ANSYS Fluent

Abstract

The Bypass Dual Throat Nozzle (BDTN) is a fluidic flow-control concept that deflects an exhaust jet without mechanical actuators. This work presents my part of a team project: a two-dimensional CFD study of how the bypass injection angle and the cavity convergence angle affect thrust vectoring. Steady-state RANS simulations were run in ANSYS Fluent with the realizable $k-\epsilon$ turbulence model for Nozzle Pressure Ratios (NPR) of 2 to 8, bypass angles from 30° to 120° and cavity convergence angles of 30° , 50° and 60° . The largest jet deflection of 27.8° occurred at $\text{NPR} = 2$ with a 45° bypass angle and a 50° convergence angle. Vectoring decreased with increasing NPR and with increasing bypass angle, and among the three convergence angles tested, 50° performed best.

Keywords: Bypass Dual Throat Nozzle, Fluidic Thrust Vectoring, Bypass Injection Angle, Cavity Convergence Angle, ANSYS Fluent

Methodology

Item	Setting
Geometry	2D model in ANSYS Design Modeler; inlet height 60 mm, primary throat 20 mm, bypass duct 2.6 mm, exit height 24 mm
Domain	Extended $20\times$ nozzle exit height axially and $30\times$ transversely
Mesh	Structured 2D mesh, 272,889 cells; refined near throat, cavity walls and bypass slot; grid independence verified
Solver	ANSYS Fluent, density-based implicit, steady RANS, second-order upwind
Turbulence model	Realizable $k-\epsilon$
Boundary conditions	Pressure inlet (NPR 2, 4, 6, 8), pressure outlet at ambient, adiabatic no-slip walls, air as working fluid
Parameters studied	Bypass angle 30° – 120° (measured from the upstream direction of the main flow); cavity convergence angle 30° , 50° , 60°
Convergence	Residuals below $1e-6$ and thrust coefficient stabilised
Validation	Centerline static pressure trends compared with Afridi et al. (2024); good agreement

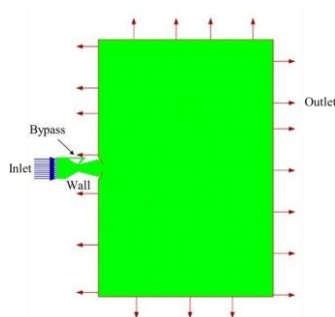


Figure 1: Computational domain and boundary conditions of the BDTN

Results: Bypass Injection Angle

- Smaller bypass angles (30° – 45°), where the bypass flow opposes the main jet, create the strongest cavity recirculation and pressure asymmetry, and give the largest deflection.
- At 45° the vectoring angle is 27.8° at $\text{NPR} = 2$ and 23.0° at $\text{NPR} = 8$. At 120° the injection is nearly parallel to the main flow, so coupling is weak and deflection is lowest.
- At low NPR the vectoring is pressure-driven; at high NPR the main jet dominates and shock cells form, so the vectoring angle falls.

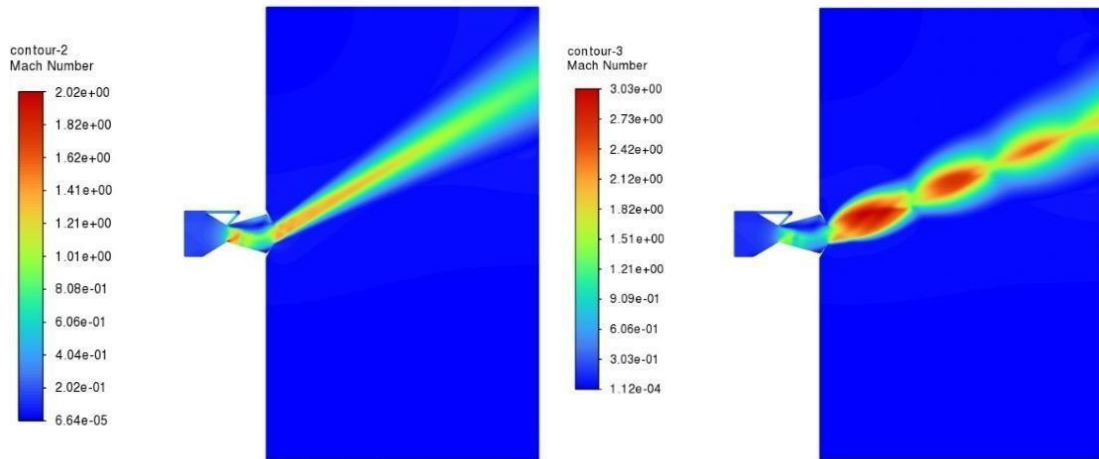


Figure 2: Mach number contours at 45° bypass angle: $\text{NPR} = 2$ (left) and $\text{NPR} = 8$ (right)

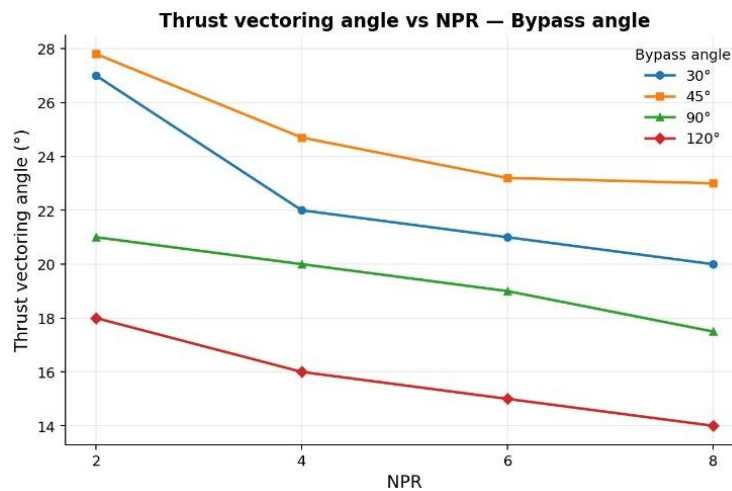


Figure 3: Variation of thrust vectoring angle with NPR for different bypass angles

Table 1: Thrust vectoring angle ($^\circ$) by bypass angle (approximate values read from Figure 3)

Bypass angle	NPR 2	NPR 4	NPR 6	NPR 8
30°	27.0	22.0	21.0	20.0
45°	27.8	24.7	23.2	23.0
90°	21.0	20.0	19.0	17.5
120°	18.0	16.0	15.0	14.0

Results: Cavity Convergence Angle

- A 30° ramp is too gentle to guide and compress the deflecting jet, giving the lowest vectoring angle (about 16° at NPR = 2).
- A 50° ramp gave the largest deflection, 27.8° at NPR = 2, and stayed highest across all NPR values.
- A steeper 60° ramp reduced the vectoring angle to about 21° at NPR = 2. This is attributed to strong flow blockage and oblique shock losses in the cavity.

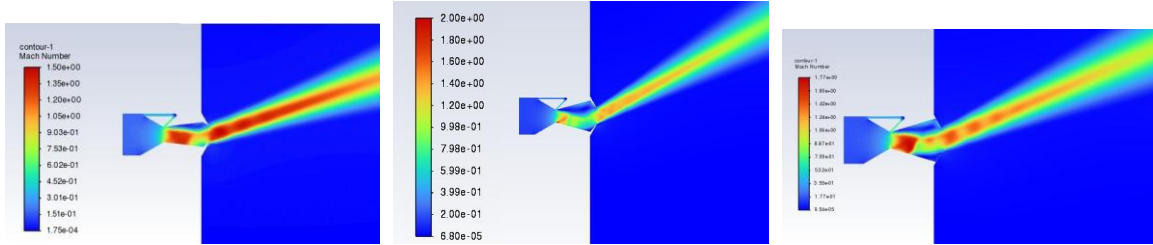


Figure 4: Mach number contours at NPR = 2, 45° bypass angle: convergence angle 30° (left), 50° (centre), 60° (right)

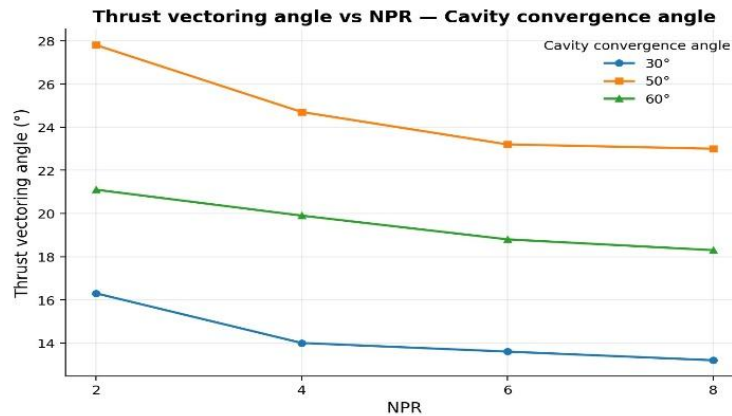


Figure 5: Variation of thrust vectoring angle with NPR for different cavity convergence angles

Table 2: Thrust vectoring angle (°) by cavity convergence angle (approximate values read from Figure 5)

Convergence angle	NPR 2	NPR 4	NPR 6	NPR 8
30°	16.3	14.0	13.6	13.2
50°	27.8	24.7	23.2	23.0
60°	21.1	19.9	18.8	18.3

Conclusion

- The maximum jet deflection of **27.8°** was obtained at NPR = 2 with a 45° bypass angle and a 50° cavity convergence angle.
- Vectoring angle decreases as NPR increases and as the bypass angle increases; the best performance is at bypass angles of 30°–45° and low NPR.
- Of the convergence angles tested (30°, 50°, 60°), 50° gave the highest vectoring angle at every NPR.
- The BDTN is an effective passive thrust vectoring concept for lightweight, maintenance-free propulsion systems.

My Role

I carried out the CFD simulations and post-processing for the bypass injection angle study and the cavity convergence angle study, including running the cases in ANSYS Fluent, preparing the Mach number contours and plotting the vectoring angle results. The cavity length study and other parts of the project were carried out by my teammates.

Reference

Afridi, S., Khan, T. A., Shah, I., Ali, Y., Mumtaz Qadri, M. N., & Li, W. (2024). Effect of bypass duct on the thrust vectoring performance of dual throat nozzle in a supersonic aircraft. *ASME Journal of Fluids Engineering*, 146(6), 061206.